Condition-Dependent Collapse Dynamics in Multi-Turn LLM Self-Play

Sohail Mohammad

February 2026

Abstract

We present a systematic empirical study of conversational collapse dynamics in multi-turn self-play between 7B-parameter language models. Using a preregistered confirmatory design (N=720 trajectories: 4 conditions $\times$ 36 seeds $\times$ 5 repeats, 40 turns each), we characterize collapse behavior across homogeneous self-play configurations (Llama-3.1-8B, Qwen2.5-7B, Mistral-7B-v0.3) and a heterogeneous rotation condition. Baseline execution achieved complete tuple closure (720/720) with full protocol integrity. Collapse dynamics were condition-dependent: Qwen2.5-7B homogeneous self-play exhibited the highest mean collapse rate (0.773), while Mistral-7B showed the lowest (0.141). Heterogeneous model rotation produced intermediate collapse (0.250). The preregistered detector reliability criterion (Cohen's $\kappa \geq 0.80$) was not met ($\kappa = 0.566$; raw agreement 92.8%). Consequently, all collapse-pattern findings are reported as descriptive rather than detector-validated.

1 Introduction

Long-horizon language-model interaction can drift into repetitive output regimes that degrade novelty and responsiveness. Related concerns about repetition, degeneration, and reliability in generative systems are well documented in single-agent and dialogue settings [Holtzman et al., 2020, Li et al., 2016, Welleck et al., 2020]. However, less is known about how such behavior distributes across controlled multi-turn self-play conditions when protocol settings are held fixed.

Paper A was designed as a preregistered baseline characterization study rather than an intervention paper. The confirmatory objective was to estimate condition-level collapse patterns under a locked detector and to evaluate detector reliability against an independent audit gate. Because the preregistered reliability threshold was not met ($\kappa = 0.566$, threshold 0.80), this manuscript follows Path B claim governance: transparent downgrade of inferential scope.

Accordingly, our claims are limited to: (i) execution closure and protocol integrity, and (ii) descriptive, condition-comparative patterns under the locked detector definition. We do not claim detector validation, causal condition effects, or mitigation efficacy.

2 What "collapse" looks like in raw transcripts

The intuition is straightforward: if two models keep talking for 40 turns, do responses stay meaningfully novel, or do they get trapped in repetitive response shells? We selected representative examples after scanning all 180 cached trajectories in the audit turn-log set.

2.1 Example A (strong repetition loop: Qwen homogeneous, HOMO_B)

Turn 18: "And there you have it! The function absolutely nails it, delivering the exact value we were hoping for ..."
Turn 24: "And there you have it! The function absolutely nails it, delivering the exact value we were hoping for ..."
Turn 39: "Got it! Here's your comment: And there you have it! The function absolutely nails it ..."

Interpretation: near-identical template reuse persists over long spans, matching the detector's collapse signature.

2.2 Example B (low-collapse trajectory: Mistral homogeneous, HOMO_C)

Turn 18: "Bias mitigation ... fairness-aware machine learning algorithms ..."
Turn 24: "Explainability and transparency are also important for regulatory compliance ..."
Turn 39: "Thank you ... don't hesitate to reach out if you have other questions ..."

Interpretation: still topically related, but content continues to move and reframe rather than locking into one repeated template.

2.3 Example C (intermediate pattern: heterogeneous rotation, HETERO_ROT)

Turn 18: "Develop and implement policies and regulations to reduce greenhouse gas emissions ..."
Turn 30: "By working together, we can make a significant difference ..."
Turn 39: "I hope this document provided a useful framework ..."

Interpretation: thematic repetition is present, but template-locking is weaker than in the highest-collapse homogeneous runs.

3 Methods

3.1 Confirmatory design

The locked confirmatory design was $4 \times 36 \times 5$: 4 conditions (HOMO_A, HOMO_B, HOMO_C, HETERO_ROT) $\times$ 36 seed prompts $\times$ 5 repeats, yielding 720 unique trajectories. Each trajectory ran for exactly 40 turns with no early stopping. The trajectory unit is a tuple (condition $\times$ seed $\times$ repeat).

3.2 Conditions and model mapping

- **HOMO_A:** Llama-3.1-8B-Instruct (rev `0e9e39f2`)
- **HOMO_B:** Qwen2.5-7B-Instruct (rev `a09a3545`)
- **HOMO_C:** Mistral-7B-Instruct-v0.3 (rev `c170c708`)
- **HETERO_ROT:** round-robin rotation across all three model families

Generation settings were fixed at temperature 0.7, top-p 0.95, and max tokens 256.

3.3 Collapse detector (locked operating point)

Collapse was detected with an embedding-based rule set using `sentence-transformers/all-MiniLM-L6-v2` [Reimers and Gurevych, 2019]. At turn t, periodicity features were defined as $s_{1,t} = \cos(e_t, e_{t-1})$ and $s_{2,t} = \cos(e_t, e_{t-2})$, with thresholds $s_{1,t} \geq 0.92$ or $s_{2,t} \geq 0.90$ sustained for at least three consecutive turns. Drift constraints were $d_1 \leq 0.08$, $d_2 \leq 0.10$, where $d_k = 1 - \cos(\cdot)$. A turn was labeled collapsed only when periodicity and low-drift criteria co-occurred.

3.4 Reliability gate protocol

The preregistered reliability gate required Cohen's $\kappa \geq 0.80$ on a stratified 180-window audit sample. Two independent LLM raters labeled windows under a locked rubric. Final gate result was **NOT MET**: $\kappa = 0.566$ (167/180 agreement; 92.8% raw agreement).

3.5 Artifacts and reproducibility contract

The canonical source artifact for confirmatory summaries is the frozen baseline matrix (coverage matrix hash recorded in the submission reproducibility index and freeze note). Tables and figures in this manuscript were generated from that frozen artifact.

4 Results

4.1 Baseline completion

The confirmatory baseline matrix completed in full: 720/720 tuples succeeded with protocol integrity (all 40 turns, no early stopping).

4.2 Condition-level collapse patterns

Table 1: Condition-level collapse summary (N=180 each).

Condition	Mean	Median	SD	N
HOMO_A (Llama)	0.457	0.475	0.311	180
HOMO_B (Qwen)	0.773	0.850	0.210	180
HOMO_C (Mistral)	0.141	0.025	0.208	180
HETERO_ROT	0.250	0.138	0.281	180

4.3 Detector reliability

The preregistered reliability criterion ($\kappa \geq 0.80$) was not met. Although raw agreement was high (92.8%), estimated reliability remained at $\kappa = 0.566$ under high collapse prevalence in the audit set.

5 Discussion

Under fixed protocol settings, collapse dynamics varied substantially by condition. The observed ordering (HOMO_B > HOMO_A > HETERO_ROT > HOMO_C) was stable in this dataset and provides a useful descriptive baseline for follow-on work.

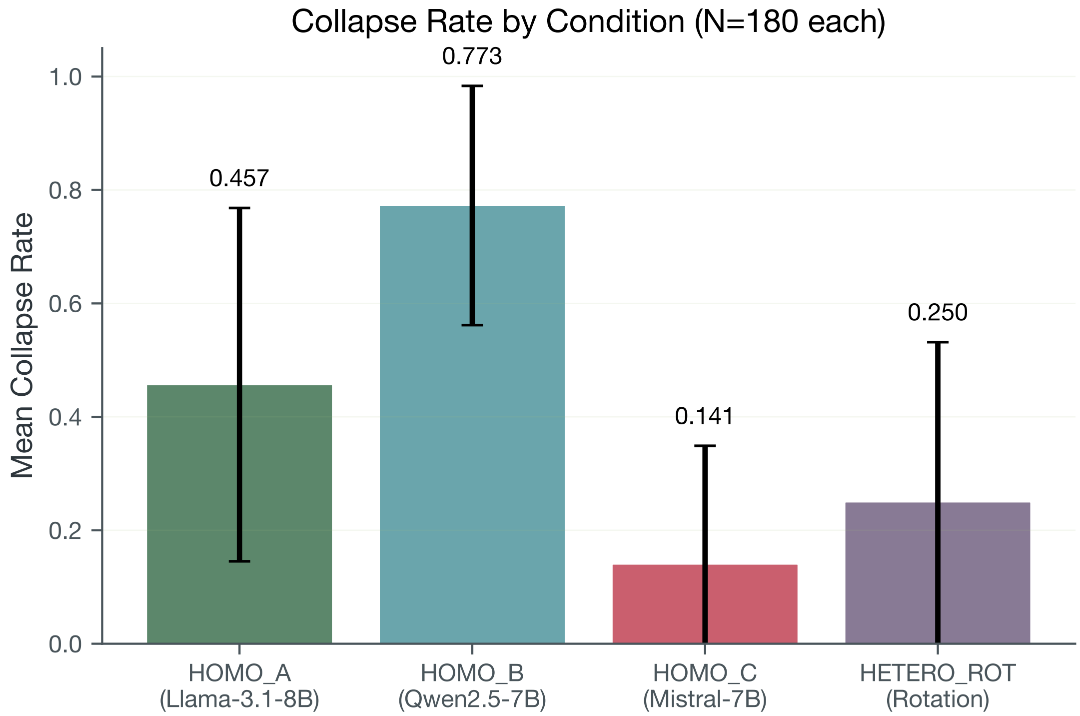

Figure 1: Mean collapse rate by condition with standard deviation error bars.

Path B constraints are central to interpretation. Because the preregistered reliability gate was not met, these results should be used as detector-defined descriptive evidence only. They support prioritizing stronger human-labeled reliability procedures and alternative detector formulations before any confirmatory or causal claim expansion.

6 Limitations

The preregistered detector reliability gate was not met ($\kappa = 0.566$; threshold 0.80). Three rubric iterations were conducted (v1.0: $\kappa = 0.011$; v2.1: $\kappa = 0.519$; v3.0: $\kappa = 0.566$). The gap between raw agreement (92.8%) and κ is consistent with strong prevalence skew. A prevalence-adjusted statistic (PABAK $= 0.856$) is reported as supplementary sensitivity information, not as gate replacement.

Consequently, all findings are descriptive and condition-comparative rather than detector-validated.

7 Conclusion

We provide a fully closed confirmatory baseline (720/720 tuples) for condition-dependent collapse behavior in 7B-model self-play under a locked protocol. The present contribution is a reproducible descriptive benchmark under Path B claim governance; detector-validation and intervention efficacy remain open tasks for future work.

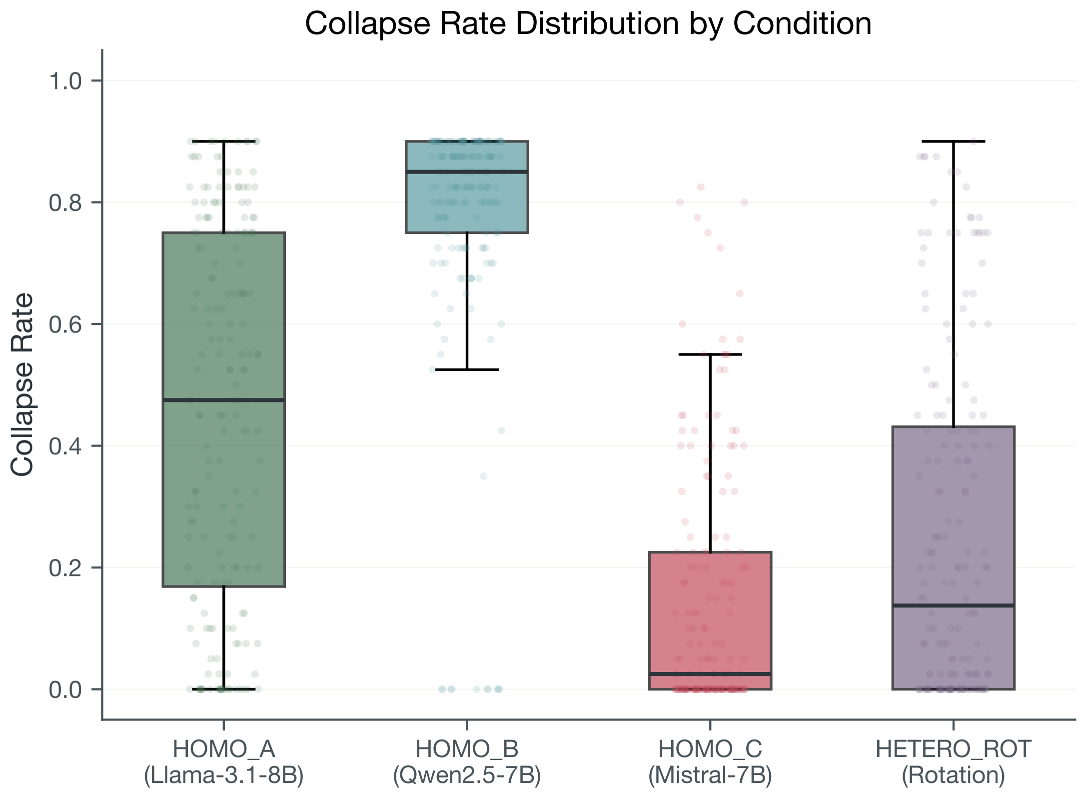

Figure 2: Collapse rate distribution by condition (boxplot with jittered strip overlay).

References

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1904.09751.

Jiwei Li, Will Monroe, and Dan Jurafsky. A diversity-promoting objective function for neural conversation models. In *Proceedings of NAACL-HLT*, pages 110–119, 2016. doi: 10.18653/v1/N16-1014.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992, 2019. doi: 10.18653/v1/D19-1410.

Sean Welleck, Ilia Kulikov, Stephen Roller, Emily Dinan, Kyunghyun Cho, and Jason Weston. Neural text generation with unlikelihood training. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1908.04319.